

Introduction

See Tong et al (2025).

Data

Unpublished work. See Tong et al (2025) for cyclone compositing, CCFs, data structure, PCs, and more. All CCFs were standardized to zero mean and unit variance prior to training. Grid-averaged albedo data is obtained from CERES SYN1deg and filtered by high cloud fraction less than 10%. The effects of solar angles on albedo are removed following the method of Cronin (2014) to better capture the contribution from variations in clouds. This solar angle corrected low cloud albedo (a_n) approximates the albedo of a given grid if the sun were overhead.

Methods

To balance predictive performance and physical interpretability, we trained a 1-layer MLP with a hyperbolic tangent (tanh) activation function on CCFs to predict a_n . The tanh activation was chosen because a_n is a continuous variable with natural saturation toward both extremes, which tanh can represent smoothly while remaining analytically interpretable. We deliberately avoid deeper networks to limit spurious interactions among CCFs. To avoid data leakage across time, the data was split chronologically, with the first ~70% of samples used for training, the subsequent ~15% for validation, and the final ~15% reserved for testing. The number of hidden neurons was varied (3 to 50) to assess the sensitivity of model performance to underfitting and overfitting. To further control overfitting, ridge (L2) regularization and early stopping based on validation loss were applied. The learning rate was fixed at 10^{-4} for all experiments.

Model predictions of a_n were evaluated on the held-out test set and compared against observations, by RMSE, R^2 , and the slope between predicted and observed a_n . The performance of the MLP was benchmarked against a linear regression model trained on principal components of the CCFs (hereafter referred to as the PC-linear model). To investigate potential nonlinear relationships between a_n and individual CCFs, partial dependence was estimated for each predictor by varying one CCF while holding the remaining CCFs fixed at their mean values. In parallel, empirical relationships were estimated by binning the data according to the rank of each CCF and computing bin-averaged values of a_n . Linear fits to the bin-averaged data were compared with the MLP-estimated partial dependence curves to assess if and by how much nonlinear effects improved predictive skill relative to linear assumptions.

Local sensitivities, representing local linearization of the MLP and capture state-dependent responses, were further quantified by computing the gradient $\partial a_n / \partial \text{CCF}$ via automatic differentiation, allowing assessment of state-dependent responses across the range of each CCF.

Finally, to probe the physical meaning of the learned representations, hidden-layer neurons were examined individually. We analyzed the weights connecting each CCF to each neuron, as well as each neuron's contribution to the output, approximated as

$$\text{contrib}_k = v_k h_k,$$

where h_k denotes the activation of neuron k and v_k denotes its weight in the output layer. This analysis provides insight into how nonlinear combinations of CCFs contribute to variability in the predicted a_n .

Results

The PC-linear model yields a normalized root-mean-square error (RMSE, normalized by the mean observed a_n) of 18% and an R^2 of 0.56 (Tong et al., 2026, manuscript under preparation). The predictions systematically overestimates lower values and underestimates higher values, resulting in a regression slope of 2.6 between predicted and observed values.

When applying the MLP directly to the CCFs, both the regression slope and the normalized RMSE remain comparable to those of the PC-linear model (Figure 1a). The MLP does not exhibit improved predictive skill for extreme values either (Figure 1b). Model performance is largely insensitive to the number of hidden neurons tested (ranging from 3 to 50) and to whether the MLP is trained on raw CCFs or on PCs derived from the CCFs (not shown). Unless otherwise noted, results shown correspond to the MLP trained on CCFs with 10 hidden neurons.

Partial dependence analysis reveals that SST, estimated inversion strength (EIS), surface wind speed (U_{10}), and temperature advection (T_{adv}), which have stronger control on a_n compared to others, exhibit predominantly linear relationships with a_n , with only weak nonlinear deviations (Figure 2). In contrast, the marine cold air outbreaks index (M), surface divergence (∇_0), and insolation display more pronounced nonlinear behavior but exert a comparatively weaker influence on a_n . Applying the MLP to PCs yields consistent results: a_n is most sensitive to variations in PC2, and this relationship is largely linear (Figure 3).

To examine how the MLP internally represents the CCFs, we analyzed the hidden-layer neurons. The neuron contributing most strongly to a_n is dominated by insolation (Figure 4). This is reasonable because insolation is not part of the cyclone system, and is also our motivation to treat insolation as an independent predictor in addition to the PCs in the paper (Tong et al., 2025). Although most other neurons combine multiple CCFs, model performance and sensitivity analyses indicate that interactions among predictors play a limited role in controlling a_n . These combinations also do not readily correspond to clearly interpretable physical modes without more extensive analysis.

Discussions and conclusions

Our analysis of nonlinearity can be limited, as partial dependence assumes independence among CCFs, and sensitivities are local derivatives. Therefore, nonlinear interactions could still exist. However, they are likely secondary given their limited impact on prediction skill and that the experiment on PCs, which are orthogonal, has similar results.

The lack of improvement from nonlinear models suggests that the dominant meteorological controls on a_n are well approximated by additive, quasi-linear relationships. In this project, the neural network serves primarily as a diagnostic tool rather than a performance-oriented model. In predicting low cloud albedo using CCFs in extratropical cyclones, model interpretability and physically informed predictors appear more important than model complexity.

Reference

- Cronin, T. W., 2014: On the Choice of Average Solar Zenith Angle. *J. Atmos. Sci.*, **71**, 2994–3003, <https://doi.org/10.1175/JAS-D-13-0392.1>.
- Tong, S., Wood, R., & Yuan, T. (2025). Predicting the frequency of low cloud mesoscale morphologies in Southern Ocean extratropical cyclones using cloud controlling factors. *Journal of Geophysical Research: Atmospheres*, 130, e2024JD043248. <https://doi.org/10.1029/2024JD043248>